

Unified Healthcare Intelligence Platform

Port Said Governorate — Egypt

DATABASE LOGICAL DESIGN DOCUMENT



Version 2.0 | May 2025 | Graduation Project — Business Intelligence & Fraud Detection

1. Introduction

This document presents the Logical Database Design for the Unified Healthcare Intelligence Platform (UHIP), a synthetic healthcare dataset built to represent the healthcare ecosystem of Port Said Governorate, Egypt. The design covers 25 relational tables organized across 6 functional modules, supporting Business Intelligence, Healthcare Analytics, Insurance Claims processing, Fraud Detection, Operational Analysis, and Machine Learning.

1.1 Design Principles

- **Referential Integrity** — all foreign keys reference valid primary keys in parent tables.
- **Normalization** — tables follow 3NF; lookup/reference tables are separated from transactional tables.
- **Clinical Accuracy** — diagnosis codes, procedure codes, and drug assignments follow real clinical logic.
- **Analytical Completeness** — the schema supports star-schema data warehouse modelling with visits as the central fact table.

1.2 Table Summary

Table Name	Description	Module
patients	Master patient registry with demographics	Patient Management
visits	All hospital encounters (central fact table)	Patient Management
diagnoses	ICD-style diagnosis reference catalog	Patient Management
medical_records	Clinical notes per completed visit	Patient Management
visit_procedures	Procedures performed per visit	Patient Management
prescriptions	Prescription headers per visit	Patient Management
prescription_items	Individual drug items per prescription	Patient Management
hospitals	Healthcare facility master list	Hospital Resources
departments	Clinical departments per hospital	Hospital Resources
doctors	Physician registry with specialties	Hospital Resources
doctor_schedules	Daily shift records per doctor	Hospital Resources
beds	Physical bed inventory snapshot	Hospital Resources
icu_status	ICU occupancy time-series snapshots	Hospital Resources
referrals	Inter-hospital patient transfer records	Hospital Resources
drugs	Medication catalog from Egyptian market	Pharmacy & Inventory
drug_inventory	Current drug stock levels per hospital	Pharmacy & Inventory
drug_transactions	Pharmacy movement log (purchases, dispensing)	Pharmacy & Inventory
suppliers	Pharmaceutical supplier directory	Pharmacy & Inventory
claims	Insurance claim submissions	Insurance Claims
claim_items	Itemised procedures and drugs per claim	Insurance Claims
claim_approvals	Insurer review decisions per claim	Insurance Claims
procedures	Medical procedure catalog with costs	Procedures Reference
patient_feedback	Patient satisfaction survey responses	Citizen Services

2. Module Descriptions

Module: Patient Management

The core module. The patients table is the master registry. visits is the central fact table that connects patients, hospitals, doctors, departments, and diagnoses. Every clinical activity (medical records, procedures, prescriptions) references a visit.

Module: Hospital Resource Management

Covers the physical and human infrastructure: hospitals, departments, doctors, shift schedules, bed inventory, ICU occupancy snapshots, and inter-hospital referrals. Enables capacity planning, staffing analysis, and geographic access studies.

Module: Pharmacy & Inventory

Manages the drug catalog, per-hospital stock levels, movement transactions (purchases, dispensing, wastage, returns), and supplier records. Suspicious dispensing patterns without matching prescriptions are embedded as drug theft fraud signals.

Module: Insurance Claims

Records every claim submission linked to a visit, with itemised procedure and drug billing. The claim_approvals table captures the insurer review decision including approval status and rejection reasons.

Module: Citizen Services

Captures post-visit patient satisfaction feedback with numeric rating (1–5) and free-text comments. Enables hospital performance benchmarking and waiting-time vs satisfaction correlation analysis.

3. Table Definitions

Each table below shows all columns with data type, key constraint, description, and a sample value. Green rows = Primary Key columns. Orange rows = Foreign Key columns.

PK	Primary Key	Uniquely identifies each row. Never NULL, never duplicate.
FK	Foreign Key	References the PK of a parent table. Enforces referential integrity.

Module 1 — Patient Management

patients

Master registry of all patients. The primary dimension table — referenced by virtually every other table.

Column Name	Data Type	Key	Description	Sample Value
patient_id	VARCHAR(12)	PK	Unique patient identifier. Format: PAT + 6 digits.	PAT000001
national_id	VARCHAR(14)		Egyptian National ID. Encodes birth date, governorate (11=Port Said), sequence, check digit.	296091911992646
first_name	VARCHAR(50)		Patient first name (Egyptian Arabic transliterated).	Reda
last_name	VARCHAR(50)		Patient family name.	Madkour
gender	VARCHAR(6)		Biological sex. Values: Male, Female.	Male
birth_date	DATE		Date of birth. Range: 1945-01-01 to 2025-05-09. Young-skewed distribution.	1996-09-19
phone	VARCHAR(12)		Egyptian mobile number (010/011/012/015 prefix). ~2% NULL (data quality simulation).	01105296193
address	VARCHAR(150)		Street address. Linked to primary hospital district in Port Said.	24 Mohamed Mahmoud St, El Sharq, Port Said
governorate	VARCHAR(20)		Always "Port Said". All patients are local to the study area.	Port Said
blood_type	VARCHAR(3)		ABO/Rh blood group. Values: A+, A-, B+, B-, AB+, AB-, O+, O-.	AB+
emergency_contact	VARCHAR(12)		Emergency contact phone. ~4% NULL.	01111493512
created_at	DATETIME		System registration timestamp. Always before the study period start (2024-05-09).	2018-03-08 14:05:00

diagnoses

Reference catalog of 120 ICD-style diagnoses across 14 disease categories. Used as a dimension in visits.

Column Name	Data Type	Key	Description	Sample Value
diagnosis_code	VARCHAR(4)	PK	ICD-like code. Format: D + 3 digits (D001–D120).	D001

Column Name	Data Type	Key	Description	Sample Value
diagnosis_name	VARCHAR(100)		Full clinical diagnosis name. Examples: Essential Hypertension, Acute Myocardial Infarction.	Essential Hypertension
diagnosis_category	VARCHAR(30)		Disease system. Values: Cardiovascular, Respiratory, Endocrine, Gastrointestinal, Neurological, Urological, Musculoskeletal, Trauma, Infectious, Hematological, Mental Health, Skin, ENT, Pediatric.	Cardiovascular
severity_level	VARCHAR(10)		Clinical severity. Values: Mild, Moderate, Severe, Critical, Chronic.	Chronic

visits

Central FACT TABLE. Every patient-hospital encounter. Connects patients, hospitals, doctors, departments, and diagnoses.

Column Name	Data Type	Key	Description	Sample Value
visit_id	VARCHAR(12)	PK	Unique visit ID. Format: VIS + 8 digits.	VIS00000001
patient_id	VARCHAR(12)	FK	Visiting patient. → patients.patient_id	PAT000001
hospital_id	VARCHAR(4)	FK	Facility of visit. → hospitals.hospital_id	H001
doctor_id	VARCCHAR(7)	FK	Attending physician. → doctors.doctor_id	DOC0080
department_id	VARCHAR(8)	FK	Treating department. → departments.department_id	DEPT0013
visit_date	DATE		Date of encounter. Range: 2024-05-09 to 2025-05-09. Winter peak months (Jan–Mar).	2024-08-13
visit_type	VARCHAR(15)		Encounter type. Values: Emergency, Outpatient, Inpatient, Follow-up, Routine Check.	Outpatient
diagnosis_code	VARCHAR(4)	FK	Primary diagnosis. → diagnoses.diagnosis_code	D023
symptoms	VARCHAR(100)		Free-text symptom description. 29 rotating clinical templates.	muscle cramps and weakness
visit_status	VARCHAR(10)		Outcome. Values: Completed (~85%), No Show (~8%), Cancelled (~7%).	Completed
waiting_time	INTEGER		Minutes waited before being seen. Govt hospitals: mean 85 min. Private: mean 25 min. NULL if not Completed.	77
total_cost	DECIMAL(10,2)		Total billed amount (EGP). Inpatient/Emergency ~3x Outpatient. NULL if not Completed.	902.62

Foreign Key Relationships:

- patient_id → patients.patient_id
- hospital_id → hospitals.hospital_id
- doctor_id → doctors.doctor_id
- department_id → departments.department_id
- diagnosis_code → diagnoses.diagnosis_code

medical_records

Clinical documentation for each completed visit. One record per completed visit. Contains physician notes, treatment plan, and follow-up decision.

Column Name	Data Type	Key	Description	Sample Value
record_id	VARCHAR(12)	PK	Unique record ID. Format: MR + 8 digits.	MR00000001
visit_id	VARCHAR(12)	FK	Visit being documented. → visits.visit_id. 1:1 with completed visits.	VIS00000001
procedure_code	VARCHAR(4)		Primary procedure performed. References procedures catalog.	P001
diagnosis_notes	VARCHAR(300)		Clinical narrative (vital signs, examination findings). Auto-generated realistic templates.	Elderly patient presenting with muscle cramps. BP 145/90...
treatment_notes	VARCHAR(200)		Treatment plan and management decisions.	Oral medication prescribed. Patient counselled.
follow_up_required	VARCHAR(3)		Whether follow-up is needed. Values: Yes (~32%), No (~68%).	No
created_at	DATETIME		Record creation timestamp. Same day as visit_date, between 08:00 and 22:00.	2024-08-13 16:10:00

Foreign Key Relationships:

- visit_id → visits.visit_id

visit_procedures

Junction table: procedures performed during each completed visit. Multiple procedures per visit (avg ~2 per visit). Clinically matched to diagnosis.

Column Name	Data Type	Key	Description	Sample Value
visit_procedure_id	VARCHAR(12)	PK	Unique record ID. Format: VP + 8 digits.	VP00000001
visit_id	VARCHAR(12)	FK	Visit during which procedure was performed. → visits.visit_id	VIS00000001
procedure_code	VARCHAR(4)	FK	Procedure performed. → procedures.procedure_code. Clinically appropriate for the visit diagnosis.	P001
procedure_cost	DECIMAL(10,2)		Actual billed cost (EGP). = procedures.expected_cost × (0.88–1.12). Fraud visits inflated 1.2–2.1x.	132.90
procedure_date	DATE		Date procedure performed. Always equals the parent visit_date.	2024-08-13

Foreign Key Relationships:

- visit_id → visits.visit_id
- procedure_code → procedures.procedure_code

prescriptions

Prescription header. Generated for ~75% of completed visits. Links a visit to the prescribing doctor.

Column Name	Data Type	Key	Description	Sample Value
prescription_id	VARCHAR(12)	PK	Unique prescription ID. Format: RX + 8 digits.	RX00000001
visit_id	VARCHAR(12)	FK	Visit during which prescription was written. → visits.visit_id	VIS00000001
doctor_id	VARCHAR(7)	FK	Prescribing physician. → doctors.doctor_id	DOC0080
prescription_date	DATE		Date written. Always equals the parent visit_date.	2024-08-13
notes	VARCHAR(100)		Pharmacist notes. Template: "Take as directed. Follow up in N days." ~10% empty.	Take as directed. Follow up in 90 days.

Foreign Key Relationships:

- visit_id → visits.visit_id
- doctor_id → doctors.doctor_id

prescription_items

Individual drug line items within prescriptions. Drugs are clinically matched to the visit diagnosis. Quantity = duration × daily dose frequency. ~3% have 2x quantity as fraud signal.

Column Name	Data Type	Key	Description	Sample Value
prescription_item_id	VARCHAR(12)	PK	Unique item ID. Format: PI + 8 digits.	PI00000001
prescription_id	VARCHAR(12)	FK	Parent prescription. → prescriptions.prescription_id	RX00000001
drug_id	VARCHAR(5)	FK	Drug prescribed. → drugs.drug_id. Matched to diagnosis therapeutic category.	DR105
dosage	VARCHAR(10)		Dose per administration (e.g., 500mg, 10mg, 1g).	10mg
frequency	VARCHAR(25)		Dosing schedule. Values: Once daily, Twice daily, Every 12 hours, Three times daily, Every 8 hours, Every 6 hours, As needed.	Three times daily
duration_days	INTEGER		Prescription duration. Range: 3–90 days, clinical range per drug category.	7
quantity	INTEGER		Total units dispensed = duration_days × daily_dose_count.	21

Foreign Key Relationships:

- prescription_id → prescriptions.prescription_id
- drug_id → drugs.drug_id

Module 2 — Hospital Resource Management

hospitals

Master list of 8 healthcare facilities in Port Said. The anchor dimension for all hospital-related analytics.

Column Name	Data Type	Key	Description	Sample Value
hospital_id	VARCHAR(4)	PK	Unique hospital ID. Values: H001–H008.	H001
hospital_name	VARCHAR(80)		Official facility name.	Port Said General Hospital
hospital_type	VARCHAR(15)		Classification. Values: Government, Private, Specialized, Teaching.	Government
governorate	VARCHAR(20)		Always "Port Said".	Port Said
district	VARCHAR(25)		Port Said district. Values: El Sharq, Port Fouad, El Arab, El Manakh, El Zohour, El Dawahy, Mubarak District.	El Sharq
address	VARCHAR(100)		Street address of the facility.	El Sharq St, Port Said
phone	VARCHAR(15)		Main hospital phone number.	0663241100
total_beds	INTEGER		Licensed bed capacity. Range: 80 (Heart Center) to 600 (Port Said General).	600
icu_capacity	INTEGER		Number of dedicated ICU beds. Range: 10–60.	60
Longitude	DECIMAL(9,6)		Stores the geographical longitude coordinate of the hospital or healthcare facility	30.0444
latitude	DECIMAL(9,6)		Stores the geographical latitude coordinate of the hospital or healthcare facility location.	30.0444
manager_name	VARCHAR(100)			Dr. Ahmed Hassan
manager_email	VARCHAR(150)			ahmed.hassan@hospital.com
manager_phone	VARCHAR(20)			+20 101 234 5678

departments

Clinical and support departments within each hospital. Large hospitals have 20 departments; private hospitals have 12.

Column Name	Data Type	Key	Description	Sample Value
department_id	VARCHAR(8)	PK	Unique ID. Format: DEPT + 4 digits.	DEPT0001
hospital_id	VARCHAR(4)	FK	Hospital containing this department. → hospitals.hospital_id	H001
department_name	VARCHAR(40)		Department name. Values: Emergency, ICU, Internal Medicine, Cardiology, Pulmonology, Neurology, Gastroenterology, Nephrology, Endocrinology, Orthopedics, General Surgery, Pediatrics, Obstetrics & Gynecology, Dermatology, ENT, Psychiatry, Oncology, Radiology, Laboratory, Pharmacy.	Emergency
floor_number	VARCHAR(2)		Floor location. 1=ground (critical depts), 2=medical wards, 3=specialty/surgical.	1

Foreign Key Relationships:

- hospital_id → hospitals.hospital_id

doctors

All physicians across 8 hospitals. 100% unique names. Specialty matches department assignment.

Column Name	Data Type	Key	Description	Sample Value
doctor_id	VARCHAR(7)	PK	Unique doctor ID. Format: DOC + 4 digits.	DOC0001
department_id	VARCHAR(8)	FK	Assigned department. → departments.department_id	DEPT0001
first_name	VARCHAR(50)		Egyptian first name.	Basel
last_name	VARCHAR(50)		Egyptian family name.	Ali
specialty	VARCHAR(40)		Medical specialty matching department. 20 specialty values aligned to department names.	Emergency Medicine
years_experience	INTEGER		Years in medical practice. Range: 1–38.	25
phone	VARCHAR(12)		Direct phone number.	01165206259
employment_status	VARCHAR(10)		Current status. Values: Active (~90%), On Leave (~10%).	Active

Foreign Key Relationships:

- hospital_id → hospitals.hospital_id
- department_id → departments.department_id

doctor_schedules

Daily shift records covering the full 12-month study period. Government hospital doctors show ~30% double-shift (overtime) rate.

Column Name	Data Type	Key	Description	Sample Value
schedule_id	VARCHAR(11)	PK	Unique shift record. Format: SCH + 7 digits.	SCH0000001
doctor_id	VARCHAR(7)	FK	Assigned doctor. → doctors.doctor_id	DOC0001

Column Name	Data Type	Key	Description	Sample Value
shift_date	DATE		Calendar date of shift. Range: 2024-05-09 to 2025-05-09.	2024-05-09
shift_start	VARCHAR(5)		Shift start time. Values: 08:00, 14:00, 20:00.	20:00
shift_end	VARCHAR(5)		Shift end time. Values: 14:00, 20:00, 08:00 (overnight).	08:00

Foreign Key Relationships:

- doctor_id → doctors.doctor_id

beds

Physical bed inventory snapshot across all hospitals. Types: ICU, Emergency, Standard ward beds.

Column Name	Data Type	Key	Description	Sample Value
bed_id	VARCHAR(9)	PK	Unique bed ID. Format: BED + 6 digits.	BED000001
department_id	VARCHAR(8)	FK	Department location. → departments.department_id	DEPT0002
bed_number	VARCHAR(10)		Human-readable label. Format: [TYPE]-[NNN]. Examples: ICU-001, ER-005, INT-012.	ICU-001
bed_type	VARCHAR(10)		Category. Values: ICU (65% occupied), Emergency (70% occupied), Standard (55% occupied).	ICU
availability_status	VARCHAR(20)		Current status. Values: Occupied, Available, Under Maintenance (~5%).	Occupied

Foreign Key Relationships:

- hospital_id → hospitals.hospital_id
- department_id → departments.department_id

icu_status

Time-series ICU occupancy snapshots recorded every 3 days. Enables seasonal ICU demand analysis.

Column Name	Data Type	Key	Description	Sample Value
icu_status_id	VARCHAR(9)	PK	Snapshot ID. Format: ICU + 6 digits.	ICU000001
hospital_id	VARCHAR(4)	FK	Hospital measured. → hospitals.hospital_id	H001
occupied_beds	INTEGER		ICU beds in use. Winter peak multiplier 1.7–1.8x. Summer low 0.85x.	40
available_beds	INTEGER		Unoccupied ICU beds. = icu_capacity – occupied_beds.	20
update_time	DATETIME		Snapshot timestamp. Random time between 06:00 and 22:00.	2024-05-09 10:16:00

Foreign Key Relationships:

- hospital_id → hospitals.hospital_id

referrals

Inter-hospital patient transfer records. Linked to a source completed visit. Reasons include capacity overflow and specialty shortages.

Column Name	Data Type	Key	Description	Sample Value
referral_id	VARCHAR(9)	PK	Unique referral ID. Format: REF + 6 digits.	REF000001
patient_id	VARCHAR(12)	FK	Transferred patient. → patients.patient_id	PAT049558
from_hospital_id	VARCHAR(4)	FK	Sending hospital. → hospitals.hospital_id	H004
to_hospital_id	VARCHAR(4)	FK	Receiving hospital. Always ≠ from_hospital_id. → hospitals.hospital_id	H008
referral_reason	VARCHAR(80)		Transfer reason. 12 values: Capacity exceeded, ICU unavailable, Specialty not available, Surgical intervention required, etc.	ICU bed unavailable - critical patient
referral_date	DATE		Date of referral. Equals the source visit date.	2025-03-31
referral_status	VARCHAR(10)		Outcome. Values: Completed (60%), Pending (30%), Cancelled (10%).	Completed

Foreign Key Relationships:

- patient_id → patients.patient_id
- from_hospital_id → hospitals.hospital_id
- to_hospital_id → hospitals.hospital_id

Module 3 — Pharmacy & Inventory

drugs

Medication reference catalog sourced from the Egyptian pharmaceutical market. 175 unique drugs across 21 therapeutic categories.

Column Name	Data Type	Key	Description	Sample Value
drug_id	VARCHAR(5)	PK	Unique drug ID. Format: DR + 3 digits (DR001–DR175).	DR001
drug_name	VARCHAR(80)		Brand/trade name with strength and formulation (e.g., Zytiga 500 mg 60 tab).	Zytiga 500 mg 60 tab.
generic_name	VARCHAR(60)		International non-proprietary name (INN/generic).	Abiraterone
manufacturer	VARCHAR(50)		Egyptian manufacturer or local subsidiary. 26 manufacturers including EIPICO, Pfizer Egypt, Novartis Egypt.	Pfizer Egypt
drug_category	VARCHAR(30)		Therapeutic class. 21 values: Antibiotic, Antidiabetic, Antihypertensive, Analgesic/NSAID, Lipid-Lowering, Cardiac, Respiratory, Corticosteroid, Psychotropic/CNS, Oncology, etc.	Oncology
unit_price	DECIMAL(8,2)		Price per unit (EGP). Range: EGP 5 (Paracetamol) to EGP 1,146 (Abiraterone/oncology). IV drugs priced higher.	1145.93

drug_inventory

Current stock levels: 8 hospitals × 175 drugs = 1,400 records. Point-in-time snapshot. Includes stockouts (~7%) and near-expiry batches (~5%).

Column Name	Data Type	Key	Description	Sample Value
inventory_id	VARCHAR(9)	PK	Unique record ID. Format: INV + 6 digits.	INV000001
hospital_id	VARCHAR(4)	FK	Holding hospital. → hospitals.hospital_id	H001
drug_id	VARCHAR(5)	FK	Stocked drug. → drugs.drug_id	DR001
quantity_available	INTEGER		Units in stock. Near-stockout: 0–5 units (~7%). Normal: 20–500. Alert when < reorder_level.	140
reorder_level	INTEGER		Minimum stock threshold triggering replenishment. Range: 10–50.	40
expiration_date	DATE		Batch expiry date. Near-expiry: within 60 days of study end (~5% of records).	2026-06-02
last_updated	DATETIME		Last inventory update timestamp. Within 30 days before 2025-05-09.	2025-04-25 13:06:00

Foreign Key Relationships:

- hospital_id → hospitals.hospital_id
- drug_id → drugs.drug_id

drug_transactions

Pharmacy movement log (350 transactions × 8 hospitals). Records all stock movements: purchases received, drugs dispensed, wastage, returns, and adjustments.

Column Name	Data Type	Key	Description	Sample Value
transaction_id	VARCHAR(9)	PK	Unique transaction ID. Format: DT + 7 digits.	DT0000001
drug_id	VARCHAR(5)	FK	Drug involved. → drugs.drug_id	DR022
hospital_id	VARCHAR(4)	FK	Hospital location. → hospitals.hospital_id	H001
transaction_type	VARCHAR(12)		Movement type. Values: Purchase (20%), Dispensing (50%), Wastage (10%), Return (10%), Adjustment (10%).	Purchase
quantity	INTEGER		Units moved. Purchase: 50–500. Normal dispensing: 1–50. Wastage/Return/Adjustment: 1–30.	149
transaction_date	DATE		Date of transaction. Range: 2024-05-09 to 2025-05-09.	2024-07-05
performed_by	VARCHAR(60)		Staff member name who performed the transaction.	Hind Shalaby

Foreign Key Relationships:

- drug_id → drugs.drug_id
- hospital_id → hospitals.hospital_id

suppliers

Pharmaceutical supplier directory for Port Said region. Used for procurement management and supply chain analytics.

Column Name	Data Type	Key	Description	Sample Value
supplier_id	VARCHAR(6)	PK	Unique supplier ID. Values: SUP001–SUP008.	SUP001
supplier_name	VARCHAR(60)		Company trading name.	Pharma Delta Co.
contact_person	VARCHAR(50)		Primary contact name and title.	Eng. Hisham Badawi
phone	VARCHAR(15)		Contact phone number.	0123456789
address	VARCHAR(100)		Supplier office address in Port Said or Cairo.	15 El-Nasr Road, Port Said

Module 4 — Insurance Claims

procedures

Medical procedure reference catalog. 201 procedures drawn from the Egyptian healthcare services list. Used as a dimension in visit_procedures and claim_items.

Column Name	Data Type	Key	Description	Sample Value
procedure_code	VARCHAR(4)	PK	Unique procedure ID. Format: P + 3 digits (P001–P201).	P001
procedure_name	VARCHAR(120)		Full procedure name. Examples: Complete Blood Count (CBC), MRI Brain Without Contrast, Percutaneous Coronary Intervention (PCI).	Complete Blood Count (CBC)
procedure_category	VARCHAR(30)		Clinical category. 20 values: Laboratory, Radiology, Cardiology, Cardiology Intervention, Surgery, Orthopedic, Orthopedic Surgery, Endoscopy, ICU, Nephrology, Hematology, Neurology, Pulmonology, Physiotherapy, Emergency, Nursing, Consultation, Diagnostic Assessment, Dermatology, Urology.	Laboratory
expected_cost	DECIMAL(10,2)		Standard reference price (EGP). Range: EGP 80 (Blood Glucose) to EGP 90,000 (cardiac intervention). Used as billing baseline.	150
complexity_score	INTEGER		Relative clinical complexity. Scale 1–5: 1=simple lab, 2=imaging, 3=endoscopy/moderate, 4=complex surgery, 5=ICU/major surgery.	1

claims

Insurance claim submissions. One claim per completed visit (62% of completed visits generate a claim).

Column Name	Data Type	Key	Description	Sample Value
claim_id	VARCHAR(12)	PK	Unique claim ID. Format: CLM + 8 digits.	CLM00000001
patient_id	VARCHAR(12)	FK	Insured patient. → patients.patient_id	PAT065539
visit_id	VARCHAR(12)	FK	Source visit. → visits.visit_id. 1:1 (one claim per visit).	VIS00445955
hospital_id	VARCHAR(4)	FK	Submitting hospital. → hospitals.hospital_id	H005
claim_date	DATE		Submission date. = visit_date + 0–14 days (realistic processing delay).	2025-01-27
claim_amount	DECIMAL(10,2)		Total claimed amount in EGP. Normal: visit_cost × 0.88–1.05.	582.28
approved_amount	DECIMAL(10,2)		Amount approved by insurer. Approved: 95–100%. Partial: 40–75%. Rejected: 0. Pending: NULL.	581.64
claim_status	VARCHAR(20)		Decision status. Values: Approved (~55%), Partially Approved (~20%), Rejected (~15%), Pending Review (~10%).	Approved

Foreign Key Relationships:

- patient_id → patients.patient_id
- visit_id → visits.visit_id
- hospital_id → hospitals.hospital_id

claim_items

Itemised billing lines per claim. Each row is either a procedure item OR a drug item (never both). Within each claim, procedure items appear first, then drug items.

Column Name	Data Type	Key	Description	Sample Value
claim_item_id	VARCHAR(12)	PK	Unique item ID. Format: CI + 9 digits.	CI000000001
claim_id	VARCHAR(12)	FK	Parent claim. → claims.claim_id	CLM000000001
procedure_code	VARCHAR(4)	FK	Procedure billed (NULL for drug items). → procedures.procedure_code. Clinically matched to visit diagnosis.	P010
drug_id	VARCHAR(5)	FK	Drug billed (NULL for procedure items). → drugs.drug_id	
item_cost	DECIMAL(10,2)		Line-item cost in EGP. Procedures: procedure expected_cost ± 5%. Drugs: unit_price × quantity.	1200.82
quantity	INTEGER		Units billed. Procedures: always 1. Drugs: equals the prescription item quantity.	1

Foreign Key Relationships:

- claim_id → claims.claim_id
- procedure_code → procedures.procedure_code (when not NULL)
- drug_id → drugs.drug_id (when not NULL)

claim_approvals

Insurer review decision per claim. One record per claim (1:1 with claims). Documents reviewer, decision, date, and rejection reason.

Column Name	Data Type	Key	Description	Sample Value
approval_id	VARCHAR(12)	PK	Unique approval record ID. Format: APR + 8 digits.	APR000000001
claim_id	VARCHAR(12)	FK	Reviewed claim. 1:1 with claims. → claims.claim_id	CLM000000001
reviewed_by	VARCHAR(50)		Insurer staff name. 10 rotating reviewers.	Omar Khalil
approval_status	VARCHAR(20)		Decision. Values: Approved, Partially Approved, Rejected, Pending Review. Matches claims.claim_status.	Approved
approval_date	DATE		Decision date. = claim_date + 1–30 days. NULL for Pending Review.	2025-02-04
rejection_reason	VARCHAR(80)		Populated only when Rejected. 10 reason values: Insufficient documentation, Duplicate claim detected, Pre-authorization not obtained, etc.	

Foreign Key Relationships:

- `claim_id` → `claims.claim_id`

Module 6 — Citizen Services

patient_feedback

Post-visit patient satisfaction responses. Sampled from completed visits. Ratings and comments vary by hospital type — government hospitals receive lower ratings.

Column Name	Data Type	Key	Description	Sample Value
feedback_id	VARCHAR(12)	PK	Unique feedback ID. Format: FB + 8 digits.	FB00000001
patient_id	VARCHAR(12)	FK	Submitting patient. → patients.patient_id	PAT023708
hospital_id	VARCHAR(4)	FK	Rated hospital. → hospitals.hospital_id	H003
doctor_id	VARCHAR(7)	FK	Rated doctor. → doctors.doctor_id	DOC0196
rating	INTEGER		Overall satisfaction (1–5). Government avg ~2.7. Private avg ~3.8.	2
comments	VARCHAR(200)		Qualitative feedback. Positive (≥4): professional service. Negative (≤2): overcrowding, wait times, understaffing.	Emergency department was chaotic and clearly understaffed.
feedback_date	DATE		Submission date. = visit_date + 0–7 days.	2024-06-08

Foreign Key Relationships:

- patient_id → patients.patient_id
- hospital_id → hospitals.hospital_id
- doctor_id → doctors.doctor_id

4. Complete Foreign Key Reference

All foreign key relationships in the database. Child table column → Parent table primary key.

Child Table (FK Column)	Parent Table	Parent Column	Relationship
departments.hospital_id	hospitals	hospital_id	Many departments → 1 hospital
doctors.hospital_id	hospitals	hospital_id	Many doctors → 1 hospital
doctors.department_id	departments	department_id	Many doctors → 1 department
doctor_schedules.doctor_id	doctors	doctor_id	Many schedules → 1 doctor
beds.hospital_id	hospitals	hospital_id	Many beds → 1 hospital
beds.department_id	departments	department_id	Many beds → 1 department
icu_status.hospital_id	hospitals	hospital_id	Many snapshots → 1 hospital
visits.patient_id	patients	patient_id	Many visits → 1 patient
visits.hospital_id	hospitals	hospital_id	Many visits → 1 hospital
visits.doctor_id	doctors	doctor_id	Many visits → 1 doctor
visits.department_id	departments	department_id	Many visits → 1 department
visits.diagnosis_code	diagnoses	diagnosis_code	Many visits → 1 diagnosis
medical_records.visit_id	visits	visit_id	1 record → 1 visit (1:1)
visit_procedures.visit_id	visits	visit_id	Many procedures → 1 visit
visit_procedures.procedure_code	procedures	procedure_code	Many VP rows → 1 procedure
prescriptions.visit_id	visits	visit_id	Many prescriptions → 1 visit
prescriptions.doctor_id	doctors	doctor_id	Many prescriptions → 1 doctor
prescription_items.prescription_id	prescriptions	prescription_id	Many items → 1 prescription
prescription_items.drug_id	drugs	drug_id	Many items → 1 drug
referrals.patient_id	patients	patient_id	Many referrals → 1 patient
referrals.from_hospital_id	hospitals	hospital_id	Many referrals → 1 sending hospital
referrals.to_hospital_id	hospitals	hospital_id	Many referrals → 1 receiving hospital
drug_inventory.hospital_id	hospitals	hospital_id	Many inventory → 1 hospital
drug_inventory.drug_id	drugs	drug_id	Many inventory → 1 drug
drug_transactions.drug_id	drugs	drug_id	Many transactions → 1 drug
drug_transactions.hospital_id	hospitals	hospital_id	Many transactions → 1 hospital
claims.patient_id	patients	patient_id	Many claims → 1 patient
claims.visit_id	visits	visit_id	1 claim → 1 visit (1:1)

claims.hospital_id	hospitals	hospital_id	Many claims → 1 hospital
claim_items.claim_id	claims	claim_id	Many items → 1 claim
claim_items.procedure_code	procedures	procedure_code	Item → procedure (when not NULL)
claim_items.drug_id	drugs	drug_id	Item → drug (when not NULL)
claim_approvals.claim_id	claims	claim_id	1 approval → 1 claim (1:1)
patient_feedback.patient_id	patients	patient_id	Many feedback → 1 patient
patient_feedback.hospital_id	hospitals	hospital_id	Many feedback → 1 hospital
patient_feedback.doctor_id	doctors	doctor_id	Many feedback → 1 doctor

5. Recommended Data Loading Order

Load tables in this sequence to avoid foreign key constraint violations:

Step	Tables
Step 1 — Reference Tables (no FKs)	diagnoses, procedures, drugs, suppliers
Step 2 — Core Facility	hospitals
Step 3 — Departments	departments (FK → hospitals)
Step 4 — Doctors	doctors (FK → hospitals, departments)
Step 5 — Hospital Resources	doctor_schedules (FK → doctors) beds (FK → hospitals, departments) icu_status (FK → hospitals)
Step 6 — Patients	patients (no FKs)
Step 7 — Pharmacy	drug_inventory (FK → hospitals, drugs) drug_transactions (FK → hospitals, drugs)
Step 8 — Visits	visits (FK → patients, hospitals, doctors, departments, diagnoses)
Step 9 — Visit Details	medical_records (FK → visits) visit_procedures (FK → visits, procedures)
Step 10 — Prescriptions	prescriptions (FK → visits, doctors) prescription_items (FK → prescriptions, drugs)
Step 11 — Referrals	referrals (FK → patients, hospitals × 2)
Step 12 — Claims	claims (FK → patients, visits, hospitals) claim_items (FK → claims, procedures, drugs) claim_approvals (FK → claims)
Step 13 — Citizen Services	patient_feedback (FK → patients, hospitals, doctors)